

VAIDYANATHAN P K

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WORK EXPERIENCE

Software Engineer, AI Foundations

Meta

March 2024-Present Bellevue, United States

- Took a developer-facing Gen AI assistant from 0 to 1, scaling to 20K+ external developers on Meta. Designed novel Knowledge Graph System (500K+ nodes) as retrieval/context layer — 78% precision, 88% recall on golden-set benchmarks. USPTO patent filed.
- Architected AI-ready docs/SDK pipelines over 10K+ API endpoints with vector indexes for context-aware retrieval — <200ms latency, 40% token cost reduction, accelerating first-prompt to production-agent.
- Built foundational platform infrastructure (planning, memory, orchestration, evaluation) powering agents across diverse use cases company-wide; established engineering standards and reusable component architectures adopted across teams.
- Built foundational infrastructure powering agentic coding assistants, owning end-to-end technical quality across reliability, safety, performance, and DX — 85%+ code-gen accuracy, 60% faster developer onboarding.
- Meta Connect 2025 Speaker: "Optimising Developer Experience for the Agentic AI Era" to 5,000+ developers; featured in Developer Keynote (100K+ viewers).

Agentic AI LLMs Knowledge Graphs Vector DBs RAG Evaluation Safety Developer Experience 0 to 1 SDKs & Docs

Senior Software Development Engineer

Amazon, Teams: Intelligent Shopping Assistant, ShopFlow

September 2022-March 2024 Seattle, United States

- Led 10 engineers/data scientists to build Rufus, Amazon's consumer-facing GenAI shopping assistant — 0 to 1 in a fast-moving, ambiguous environment. Tackled prompt engineering, hallucination mitigation, grounding in factual product data. Proposed at hackathon; acknowledged by CEO/S-Team.
- Led 8 engineers to build scalable ML orchestration platform (MetaFlow) for training and inference pipelines — established best practices, boosting developer productivity across 20 teams.

LLMs Consumer Product Prompt Engineering Hallucination Mitigation RAG Vector DBs Python Cross-Team Leadership

Software Development Engineer II

Amazon, Teams: Alexa Communications, Alexa Shopping ASR

September 2017 – August 2022 Seattle, United States

- Led cross-org team of 8 engineers (influence over authority) to ship ML-powered voice shopping platform for Alexa — full stack from SLU to entity resolution. Onboarded 25 enterprise clients in 6 months via a robust extensibility layer.
- Led 6 engineers to build Alexa-Alexa calling across hundreds of 1P/3P devices, scaling reliably to millions of users globally. Invented novel drop-in experience used in US hospitals during COVID-19 for patient-doctor communication.
- Led 4 engineers to build automated speech recognition system on open-source ASR engines; long-tail recognition via acoustic/language model optimization significantly improved rare-name accuracy.

NLP Spoken Language Understanding ASR Entity Resolution Reliability Accessibility Python Java Acoustic Modeling

Software Development Engineer I

Amazon, Teams: Retail Selection & Price Monitoring

June 2015 – August 2017 Chennai, India

- Built Apache SPARK batch system processing PBs of Amazon Catalog data daily — predictive models for demand forecasting and competitive analysis.
- Built multi-tenant distributed crawling framework for live price updates — extraction, deduplication, real-time sync at scale for millions of items.

Apache SPARK AWS Services Java Python Distributed Systems

EDUCATION

Bachelor of Engineering, Computer Science

Anna University, First class with Distinction

2011 – 2015 Chennai, India

Higher Secondary Education

All Angels Matriculation, 98% Aggregate, District Topper

2010 – 2011 Chennai, India

PATENTS/PUBLICATIONS

- AI-Driven Developer Experience (Meta): Patent filed for novel Knowledge Graph System enabling context-aware code assistance through structured knowledge representation and semantic retrieval (USPTO, 2024-2025).
- Voice Communication Systems (Alexa): Novel communication methodology for hands-free calling with privacy and consent management. US Patent Issued.
- Speech Recognition Optimization (Alexa): Novel approach to improve ASR accuracy for long tail and rare entity names through adaptive language modeling. US Patent Issued.
- Voice Product Disambiguation (Alexa): Novel method for disambiguating product catalog items in voice interfaces. US Patent Issued.
- Speech Error Prediction (Alexa): Predictive model for ASR error detection and correction. US Patent Pending.

ACHIEVEMENTS

- Meta Connect 2025 Speaker: Featured speaker presenting "Optimising Horizon Developer Experience for the Agentic AI Era" to global developer audience, focusing on responsible development of agentic AI systems. Team's work showcased in Connect Developer Keynote.
- Meta Inventor Award: Filed patent for novel AI-driven developer experience innovations including Knowledge Graph System architecture for safe and reliable agentic coding assistance.
- Amazon Inventor Award: Awarded 4 inventor awards for technical innovations in Alexa Spoken Language Understanding, Neural LLMs, and speech recognition systems.
- Amazon DCCP Star Award: Recognized as best performer in organization for outstanding technical contributions and engineering excellence (2017).
- ACM ICPC Regional Finalist: Top 10 at Kharagpur 2014, Top 25 at Amritapuri 2015.
- Codechef Certified DSA Expert: Grade 100% in Proctored Exam.

LEADERSHIP

- Amazon Leadership Team - Management Leadership For Tomorrow: Represented Amazon to deliver tech talks for 1000 underrepresented students across the US.
- Amazon SDE Bootcamp Instructor: Instructor for Amazon SDE bootcamp to mentor new hire SDEs about Amazon culture and tech platforms.
- Education & Volunteering: Active coordinator for Amazon Future Engineer programs and served as senior volunteer for educational programs in Tamil-Nadu, India, supporting underrepresented communities.